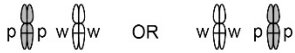


Question	Marking Guidance	Mark	Comments
06.1	1. Wash hands to remove/kill microbes OR Wear gloves/apron to prevent contamination; 2. Burning Bunsen close by to create upward current of air; 3. Disinfect bench/work on disinfected cloth to kill microbes/prevent contamination; 4. Flame instrument/equipment to sterilise/kill microbes/prevent contamination; 5. Lift lid slightly to prevent entry of microbes;	3 max	Each must include an explanation 3. Accept use sterile/antiseptic/bleach for 'disinfect' 4. Accept autoclave/use alcohol/steam/boil for flame 4. Accept scalpel/inoculating loop/wire loop/loop/forceps/spreader for instrument 1. 3. 4. & 5. Accept microorganisms/spores/bacteria/fungi/contaminants for 'microbes'
06.2	Mitosis given first 1. One division, two divisions in meiosis; 2. (Daughter) cells genetically identical, daughter cells genetically different in meiosis; 3. Two cells produced, (usually) four cells produced in meiosis; 4. Diploid to diploid/haploid to haploid, diploid to haploid in meiosis; 5. Separation of homologous chromosomes only in meiosis; 6. Crossing over only in meiosis; 7. Independent segregation only in meiosis;	2 max	Differences must be given as comparisons Ignore references to asexual / sexual reproduction, growth, repair & replacement 2. Reference to 'genetically' needed once 4. Accept same number chromosomes in mitosis, but half the number in meiosis
06.3	28;	1	Automarking

06.4	1. Separation of homologous chromosomes (occurred) OR (Independent) segregation (occurred); 2. (Arrangement/separation/segregation of chromosomes is) random/ (almost) equally frequent (in tubes 1 and 2); 3. Crossing over occurred in tube 3/10 tubes; 4. (Crossing over) is rare/infrequent/in <u>only</u> 10 tubes;	3 max	1. Accept description of homologous chromosome movement for 'separation' 1. Accept alleles for 'chromosomes' 1. Accept as a labelled diagram of chromosomes eg  3. Accept as labelled diagram of crossing over 3. and 4. "Crossing over occurred in only 10 tubes" = 2 marks
TOTAL		9	